

Philip Hargreaves

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Summary

MSc Computer Science student at UCL with two years' nuclear R&D experience at Rolls-Royce. Working across backend (Go, PHP, Node.js), full-stack web (React/Next.js) and AI/ML (LLMs, RAG, agentic workflows).

Education

University College London

MSc Computer Science

Sept 2025 – Sept 2026

Expected Distinction

- **Modules:** Data Structures and Algorithms, Computer Architecture, Operating Systems, Research Computing with C++, Object-Oriented Programming, Machine Learning, Database Fundamentals, App Engineering

University of Nottingham

MEng Mechanical Engineering

Sept 2018 – Sept 2022

First Class Honours

Experience

Student Software Engineer (UCL IXN), Intel Corporation

May 2026 – Present

- Architecting an offline, clinical AI system running local LLMs on Intel AI PCs for NHS practitioners
- Engineering a native, low-latency C++/OpenVINO core splitting inference across Intel's NPU, iGPU and CPU
- Designing RAG that grounds clinical notes in NICE guidance via semantic search + cross-encoder reranking

Mechanical Design Engineer, Rolls-Royce

May 2023 – Sept 2025

- Developed a Python-based design toolkit using Model-View-Presenter architecture and Tkinter GUI, integrating NumPy, Pandas, and Matplotlib, reducing preliminary vessel sizing time by 30%
- Achieved a 40% life extension for a critical nuclear component by developing refined brittle fracture methodologies, securing ONR regulatory approval and >£50k in recurring operational savings
- Automated FEA post-processing using Python, decreasing manual effort and ensuring data consistency
- Validated sub-contractor deliverables as technical reviewer, verifying engineering calculations and models to ensure 100% compliance with nuclear safety case requirements

Projects

CS Events Platform, UCL Department of Computer Science

React, Next.js, TypeScript, Jest, Azure

- Delivered Next.js/React events platform built on PostgreSQL/Prisma using feature-driven architecture with full unit/integration/E2E test coverage, built for the department to replace current fragmented workflows
- Built conversational AI assistant auto-filling event forms from messages or PDFs/images via Azure GPT-4.1
- Engineered Microsoft Graph cron pipeline transforming Outlook ICS invites into draft events, with single-use, DB-hashed JWT edit links emailed to external organisers and consumed atomically in Prisma

Full-Stack Auction Platform

PHP 8.4, MySQL, JavaScript, Apache, Docker

- Created custom RESTful MVC framework in vanilla PHP 8.4 routing 34 endpoints via dependency injection and repository pattern, with a normalised 14-table MySQL schema supporting recursive category filtering
- Developed transaction-safe bidding engine with outbid detection, reserve-price checks and winner resolution
- Designed weighted recommendation system using collaborative filtering, in-memory caching and SQL CTEs

Agent-Based Ecological Simulation

C++17, OpenMP, MPI, CMake, Catch2

- Built evolutionary predator-prey simulation with behavioural state machine, genetic trait inheritance and natural population dynamics, parallelised across 4 distributed processes and multi-core OpenMP threads
- Engineered cross-process reproducibility via seed broadcast, with Catch2-verified behaviour and genetics

Skills

Programming Languages: Go, Python, C++, TypeScript, JavaScript, PHP, Java, SQL, Bash

Frameworks / Libraries: React, Next.js, Node.js, Prisma, SQLC, Goose, OpenVINO, Jest, Playwright, Pytest

Tools / Cloud: Git, Docker, Linux, Claude Code, Cursor, AWS (Cloud Practitioner Certified), Azure, PostgreSQL